

732A96/TDDE15 Advanced Machine Learning

Graphical Models

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Lecture 2: Probabilistic Inference

Contents

- ▶ Probabilistic Inference for BNs
 - ▶ Naive Solution
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- ▶ Beyond Variable Elimination

Literature

- ▶ Main source
 - ▶ Koller, D. and Friedman, N. *Probabilistic Graphical Models: Principles and Techniques*. MIT Press, 2009. Chapter 9.2-9.4.
- ▶ Additional source
 - ▶ Bishop, C. M. *Pattern Recognition and Machine Learning*. Springer, 2006. Chapter 8.

Probabilistic Inference for BNs: Naive Solution

- What is the state of a random variable X_k if a random variable X_i is observed to be in the state x_i ?

$$p(x_k | x_i) = \frac{p(x_k, x_i)}{p(x_i)} = \frac{\sum_{x \setminus \{x_i, x_k\}} p(x)}{\sum_{x \setminus x_i} p(x)}$$

- For instance, $p(d) = \sum_{a,b,c} p(a, b, c, d) = \sum_{a,b,c} p(a)p(b|a)p(c|b)p(d|c)$ for the DAG $A \rightarrow B \rightarrow C \rightarrow D$.

$$\begin{array}{l} P(a^1) \quad P(b^1 | a^1) \quad P(c^1 | b^1) \quad P(d^1 | c^1) \\ + P(a^2) \quad P(b^1 | a^2) \quad P(c^1 | b^1) \quad P(d^1 | c^1) \\ + P(a^1) \quad P(b^2 | a^1) \quad P(c^1 | b^2) \quad P(d^1 | c^1) \\ + P(a^2) \quad P(b^2 | a^2) \quad P(c^1 | b^2) \quad P(d^1 | c^1) \\ + P(a^1) \quad P(b^1 | a^1) \quad P(c^2 | b^1) \quad P(d^1 | c^2) \\ + P(a^2) \quad P(b^1 | a^2) \quad P(c^2 | b^1) \quad P(d^1 | c^2) \\ + P(a^1) \quad P(b^2 | a^1) \quad P(c^2 | b^2) \quad P(d^1 | c^2) \\ + P(a^2) \quad P(b^2 | a^2) \quad P(c^2 | b^2) \quad P(d^1 | c^2) \end{array}$$

$$\begin{array}{l} P(a^1) \quad P(b^1 | a^1) \quad P(c^1 | b^1) \quad P(d^2 | c^1) \\ + P(a^2) \quad P(b^1 | a^2) \quad P(c^1 | b^1) \quad P(d^2 | c^1) \\ + P(a^1) \quad P(b^2 | a^1) \quad P(c^1 | b^2) \quad P(d^2 | c^1) \\ + P(a^2) \quad P(b^2 | a^2) \quad P(c^1 | b^2) \quad P(d^2 | c^1) \\ + P(a^1) \quad P(b^1 | a^1) \quad P(c^2 | b^1) \quad P(d^2 | c^2) \\ + P(a^2) \quad P(b^1 | a^2) \quad P(c^2 | b^1) \quad P(d^2 | c^2) \\ + P(a^1) \quad P(b^2 | a^1) \quad P(c^2 | b^2) \quad P(d^2 | c^2) \\ + P(a^2) \quad P(b^2 | a^2) \quad P(c^2 | b^2) \quad P(d^2 | c^2) \end{array}$$

Figure 9.2 Computing $P(D)$ by summing over the joint distribution for a chain $A \rightarrow B \rightarrow C \rightarrow D$; all of the variables are binary valued.

Probabilistic Inference for BNs: Naive Solution

$$\begin{array}{cccc}
 P(a^1) & P(b^1 | a^1) & P(c^1 | b^1) & P(d^1 | c^1) \\
 + P(a^2) & P(b^1 | a^2) & P(c^1 | b^1) & P(d^1 | c^1) \\
 + P(a^1) & P(b^2 | a^1) & P(c^1 | b^2) & P(d^1 | c^1) \\
 + P(a^2) & P(b^2 | a^2) & P(c^1 | b^2) & P(d^1 | c^1) \\
 + P(a^1) & P(b^1 | a^1) & P(c^2 | b^1) & P(d^1 | c^2) \\
 + P(a^2) & P(b^1 | a^2) & P(c^2 | b^1) & P(d^1 | c^2) \\
 + P(a^1) & P(b^2 | a^1) & P(c^2 | b^2) & P(d^1 | c^2) \\
 + P(a^2) & P(b^2 | a^2) & P(c^2 | b^2) & P(d^1 | c^2)
 \end{array}$$

$$\begin{array}{cccc}
 P(a^1) & P(b^1 | a^1) & P(c^1 | b^1) & P(d^2 | c^1) \\
 + P(a^2) & P(b^1 | a^2) & P(c^1 | b^1) & P(d^2 | c^1) \\
 + P(a^1) & P(b^2 | a^1) & P(c^1 | b^2) & P(d^2 | c^1) \\
 + P(a^2) & P(b^2 | a^2) & P(c^1 | b^2) & P(d^2 | c^1) \\
 + P(a^1) & P(b^1 | a^1) & P(c^2 | b^1) & P(d^2 | c^2) \\
 + P(a^2) & P(b^1 | a^2) & P(c^2 | b^1) & P(d^2 | c^2) \\
 + P(a^1) & P(b^2 | a^1) & P(c^2 | b^2) & P(d^2 | c^2) \\
 + P(a^2) & P(b^2 | a^2) & P(c^2 | b^2) & P(d^2 | c^2)
 \end{array}$$

Figure 9.2 Computing $P(D)$ by summing over the joint distribution for a chain $A \rightarrow B \rightarrow C \rightarrow D$; all of the variables are binary valued.

$$\begin{array}{ccc}
 (P(a^1)P(b^1 | a^1) + P(a^2)P(b^1 | a^2)) & P(c^1 | b^1) & P(d^1 | c^1) \\
 + (P(a^1)P(b^2 | a^1) + P(a^2)P(b^2 | a^2)) & P(c^1 | b^2) & P(d^1 | c^1) \\
 + (P(a^1)P(b^1 | a^1) + P(a^2)P(b^1 | a^2)) & P(c^2 | b^1) & P(d^1 | c^2) \\
 + (P(a^1)P(b^2 | a^1) + P(a^2)P(b^2 | a^2)) & P(c^2 | b^2) & P(d^1 | c^2)
 \end{array}$$

$$\begin{array}{ccc}
 (P(a^1)P(b^1 | a^1) + P(a^2)P(b^1 | a^2)) & P(c^1 | b^1) & P(d^2 | c^1) \\
 + (P(a^1)P(b^2 | a^1) + P(a^2)P(b^2 | a^2)) & P(c^1 | b^2) & P(d^2 | c^1) \\
 + (P(a^1)P(b^1 | a^1) + P(a^2)P(b^1 | a^2)) & P(c^2 | b^1) & P(d^2 | c^2) \\
 + (P(a^1)P(b^2 | a^1) + P(a^2)P(b^2 | a^2)) & P(c^2 | b^2) & P(d^2 | c^2)
 \end{array}$$

Figure 9.3 The first transformation on the sum of figure 9.2

Probabilistic Inference for BNs: Naive Solution

$$\begin{array}{rcl}
 & (P(a^1)P(b^1 \mid a^1) + P(a^2)P(b^1 \mid a^2)) & P(c^1 \mid b^1) \quad P(d^1 \mid c^1) \\
 + & (P(a^1)P(b^2 \mid a^1) + P(a^2)P(b^2 \mid a^2)) & P(c^1 \mid b^2) \quad P(d^1 \mid c^1) \\
 + & (P(a^1)P(b^1 \mid a^1) + P(a^2)P(b^1 \mid a^2)) & P(c^2 \mid b^1) \quad P(d^1 \mid c^2) \\
 + & (P(a^1)P(b^2 \mid a^1) + P(a^2)P(b^2 \mid a^2)) & P(c^2 \mid b^2) \quad P(d^1 \mid c^2) \\
 \\
 & (P(a^1)P(b^1 \mid a^1) + P(a^2)P(b^1 \mid a^2)) & P(c^1 \mid b^1) \quad P(d^2 \mid c^1) \\
 + & (P(a^1)P(b^2 \mid a^1) + P(a^2)P(b^2 \mid a^2)) & P(c^1 \mid b^2) \quad P(d^2 \mid c^1) \\
 + & (P(a^1)P(b^1 \mid a^1) + P(a^2)P(b^1 \mid a^2)) & P(c^2 \mid b^1) \quad P(d^2 \mid c^2) \\
 + & (P(a^1)P(b^2 \mid a^1) + P(a^2)P(b^2 \mid a^2)) & P(c^2 \mid b^2) \quad P(d^2 \mid c^2)
 \end{array}$$

Figure 9.3 The first transformation on the sum of figure 9.2

$$\begin{array}{rcl}
 & \tau_1(b^1) & P(c^1 \mid b^1) \quad P(d^1 \mid c^1) \\
 + & \tau_1(b^2) & P(c^1 \mid b^2) \quad P(d^1 \mid c^1) \\
 + & \tau_1(b^1) & P(c^2 \mid b^1) \quad P(d^1 \mid c^2) \\
 + & \tau_1(b^2) & P(c^2 \mid b^2) \quad P(d^1 \mid c^2) \\
 \\
 & \tau_1(b^1) & P(c^1 \mid b^1) \quad P(d^2 \mid c^1) \\
 + & \tau_1(b^2) & P(c^1 \mid b^2) \quad P(d^2 \mid c^1) \\
 + & \tau_1(b^1) & P(c^2 \mid b^1) \quad P(d^2 \mid c^2) \\
 + & \tau_1(b^2) & P(c^2 \mid b^2) \quad P(d^2 \mid c^2)
 \end{array}$$

Figure 9.4 The second transformation on the sum of figure 9.2

Probabilistic Inference for BNs: Naive Solution

$$\begin{array}{rcl}
 & \tau_1(b^1) & P(c^1 | b^1) \quad P(d^1 | c^1) \\
 + & \tau_1(b^2) & P(c^1 | b^2) \quad P(d^1 | c^1) \\
 + & \tau_1(b^1) & P(c^2 | b^1) \quad P(d^1 | c^2) \\
 + & \tau_1(b^2) & P(c^2 | b^2) \quad P(d^1 | c^2) \\
 \\
 & \tau_1(b^1) & P(c^1 | b^1) \quad P(d^2 | c^1) \\
 + & \tau_1(b^2) & P(c^1 | b^2) \quad P(d^2 | c^1) \\
 + & \tau_1(b^1) & P(c^2 | b^1) \quad P(d^2 | c^2) \\
 + & \tau_1(b^2) & P(c^2 | b^2) \quad P(d^2 | c^2)
 \end{array}$$

Figure 9.4 The second transformation on the sum of figure 9.2

$$\begin{array}{rcl}
 & (\tau_1(b^1)P(c^1 | b^1) + \tau_1(b^2)P(c^1 | b^2)) & P(d^1 | c^1) \\
 + & (\tau_1(b^1)P(c^2 | b^1) + \tau_1(b^2)P(c^2 | b^2)) & P(d^1 | c^2) \\
 \\
 & (\tau_1(b^1)P(c^1 | b^1) + \tau_1(b^2)P(c^1 | b^2)) & P(d^2 | c^1) \\
 + & (\tau_1(b^1)P(c^2 | b^1) + \tau_1(b^2)P(c^2 | b^2)) & P(d^2 | c^2)
 \end{array}$$

Figure 9.5 The third transformation on the sum of figure 9.2

Probabilistic Inference for BNs: Naive Solution

$$\begin{aligned}
 & (\tau_1(b^1)P(c^1 | b^1) + \tau_1(b^2)P(c^1 | b^2)) \quad P(d^1 | c^1) \\
 + & (\tau_1(b^1)P(c^2 | b^1) + \tau_1(b^2)P(c^2 | b^2)) \quad P(d^1 | c^2) \\
 \\
 & (\tau_1(b^1)P(c^1 | b^1) + \tau_1(b^2)P(c^1 | b^2)) \quad P(d^2 | c^1) \\
 + & (\tau_1(b^1)P(c^2 | b^1) + \tau_1(b^2)P(c^2 | b^2)) \quad P(d^2 | c^2)
 \end{aligned}$$

Figure 9.5 The third transformation on the sum of figure 9.2

$$\begin{aligned}
 & \tau_2(c^1) \quad P(d^1 | c^1) \\
 + & \tau_2(c^2) \quad P(d^1 | c^2) \\
 \\
 & \tau_2(c^1) \quad P(d^2 | c^1) \\
 + & \tau_2(c^2) \quad P(d^2 | c^2)
 \end{aligned}$$

Figure 9.6 The fourth transformation on the sum of figure 9.2

- Then, instead of $p(d) = \sum_{a,b,c} p(a)p(b|a)p(c|b)p(d|c)$, we can do

$$\begin{aligned}
 p(d) &= \sum_c p(d|c)\tau_2(c) \\
 &= \sum_c p(d|c) \sum_b p(c|b)\tau_1(b) \\
 &= \sum_c p(d|c) \sum_b p(c|b) \sum_a p(a)p(b|a)
 \end{aligned}$$

- Do we gain anything ? **Yes**, the former case implies 62 operations (multiplications and additions) and the latter only 18.

Probabilistic Inference for BNs: Variable Elimination

- Let us define a factor $\phi(U)$ as a function $\phi: \text{Values}(U) \rightarrow \mathbb{R}$. Let us also define $\text{Scope}(\phi) = U$.

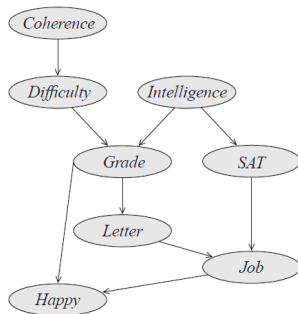


Figure 9.8 The Extended-Student Bayesian network

$$\begin{aligned} P(C, D, I, G, S, L, J, H) &= P(C)P(D | C)P(I)P(G | I, D)P(S | I) \\ &\quad P(L | G)P(J | L, S)P(H | G, J) \\ &= \phi_C(C)\phi_D(D, C)\phi_I(I)\phi_G(G, I, D)\phi_S(S, I) \\ &\quad \phi_L(L, G)\phi_J(J, L, S)\phi_H(H, G, J). \end{aligned}$$

Probabilistic Inference for BNs: Variable Elimination

- ▶ The following algorithm returns $p(y)$ where $Y = X \setminus Z$.

Algorithm 9.1 Sum-product variable elimination algorithm

Procedure Sum-Product-VE (

Φ , // Set of factors

Z , // Set of variables to be eliminated

$\prec$ // Ordering on Z

)

1 Let $Z_1, \dots, Z_k$ be an ordering of Z such that

2 $Z_i \prec Z_j$ if and only if $i < j$

3 **for** $i = 1, \dots, k$

4 $\Phi \leftarrow \text{Sum-Product-Eliminate-Var}(\Phi, Z_i)$

5 $\phi^* \leftarrow \prod_{\phi \in \Phi} \phi$

6 **return** ϕ^*

Procedure Sum-Product-Eliminate-Var (

Φ , // Set of factors

Z // Variable to be eliminated

)

1 $\Phi' \leftarrow \{\phi \in \Phi : Z \in \text{Scope}[\phi]\}$

2 $\Phi'' \leftarrow \Phi - \Phi'$

3 $\psi \leftarrow \prod_{\phi \in \Phi'} \phi$

4 $\tau \leftarrow \sum_Z \psi$

5 **return** $\Phi'' \cup \{\tau\}$

Probabilistic Inference for BNs: Variable Elimination

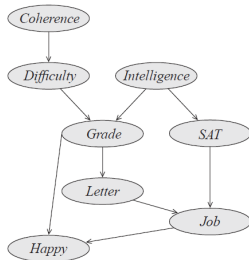


Figure 9.8 The Extended-Student Bayesian network

$$\begin{aligned}
 P(C, D, I, G, S, L, J, H) &= P(C)P(D | C)P(I)P(G | I, D)P(S | I) \\
 &\quad P(L | G)P(J | L, S)P(H | G, J) \\
 &= \phi_C(C)\phi_D(D, C)\phi_I(I)\phi_G(G, I, D)\phi_S(S, I) \\
 &\quad \phi_L(L, G)\phi_J(J, L, S)\phi_H(H, G, J).
 \end{aligned}$$

Step	Variable eliminated	Factors used	Variables involved	New factor
1	C	$\phi_C(C), \phi_D(D, C)$	C, D	$\tau_1(D)$
2	D	$\phi_G(G, I, D), \tau_1(D)$	G, I, D	$\tau_2(G, I)$
3	I	$\phi_I(I), \phi_S(S, I), \tau_2(G, I)$	G, S, I	$\tau_3(G, S)$
4	H	$\phi_H(H, G, J)$	H, G, J	$\tau_4(G, J)$
5	G	$\tau_4(G, J), \tau_3(G, S), \phi_L(L, G)$	G, J, L, S	$\tau_5(J, L, S)$
6	S	$\tau_5(J, L, S), \phi_J(J, L, S)$	J, L, S	$\tau_6(J, L)$
7	L	$\tau_6(J, L)$	J, L	$\tau_7(J)$

Table 9.1 A run of variable elimination for the query $P(J)$

Probabilistic Inference for BNs: Variable Elimination

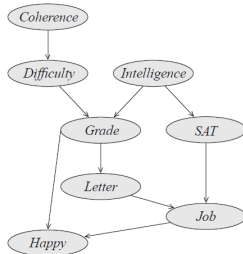


Figure 9.8 The Extended-Student Bayesian network

$$\begin{aligned}
 P(C, D, I, G, S, L, J, H) &= P(C)P(D | C)P(I)P(G | I, D)P(S | I) \\
 &\quad P(L | G)P(J | L, S)P(H | G, J) \\
 &= \phi_C(C)\phi_D(D, C)\phi_I(I)\phi_G(G, I, D)\phi_S(S, I) \\
 &\quad \phi_L(L, G)\phi_J(J, L, S)\phi_H(H, G, J).
 \end{aligned}$$

Step	Variable eliminated	Factors used	Variables involved	New factor
1	G	$\phi_G(G, I, D), \phi_L(L, G), \phi_H(H, G, J)$	G, I, D, L, J, H	$\tau_1(I, D, L, J, H)$
2	I	$\phi_I(I), \phi_S(S, I), \tau_1(I, D, L, S, J, H)$	S, I, D, L, J, H	$\tau_2(D, L, S, J, H)$
3	S	$\phi_J(J, L, S), \tau_2(D, L, S, J, H)$	D, L, S, J, H	$\tau_3(D, L, J, H)$
4	L	$\tau_3(D, L, J, H)$	D, L, J, H	$\tau_4(D, J, H)$
5	H	$\tau_4(D, J, H)$	D, J, H	$\tau_5(D, J)$
6	C	$\phi_C(C), \phi_D(D, C)$	D, J, C	$\tau_6(D)$
7	D	$\tau_5(D, J), \tau_6(D)$	D, J	$\tau_7(J)$

Table 9.2 A different run of variable elimination for the query $P(J)$

Probabilistic Inference for BNs: Variable Elimination

- What is the state of a random variable X_k if a random variable X_i is observed to be in the state x_i ?

$$p(x_k|x_i) = \frac{p(x_k, x_i)}{p(x_i)} = \frac{\sum_{x \setminus \{x_i, x_k\}} p(x)}{\sum_{x \setminus x_i} p(x)}$$

- E.g., $p(d|a) = \frac{p(d,a)}{p(a)} = \frac{\sum_{b,c} p(a)p(b|a)p(c|b)p(d|c)}{\sum_{b,c,d} p(a)p(b|a)p(c|b)p(d|c)}$ for $A \rightarrow B \rightarrow C \rightarrow D$.
- The following algorithm returns $p(y|e) = \frac{p(y,e)}{p(e)} = \frac{\phi^*}{\alpha}$ where $Y \subseteq X \setminus E$.
- Given a factor $\phi(U)$, let us define the reduced factor $\phi[E=e](Y)$ as a factor with scope $Y = U \setminus E$ such that $\phi[E=e](y) = \phi(y, z)$ where $Z = U \cap E$.

Algorithm 9.2 Using Sum-Product-VE for computing conditional probabilities

Procedure Cond-Prob-VE (

$\mathcal{K}$, // A network over $\mathcal{X}$

Y , // Set of query variables

$E = e$ // Evidence

)

- 1 $\Phi \leftarrow$ Factors parameterizing $\mathcal{K}$
- 2 Replace each $\phi \in \Phi$ by $\phi[E=e]$
- 3 Select an elimination ordering $\prec$
- 4 $Z \leftarrow \mathcal{X} - Y - E$
- 5 $\phi^* \leftarrow$ Sum-Product-VE($\Phi, \prec, Z$)
- 6 $\alpha \leftarrow \sum_{y \in \text{Val}(Y)} \phi^*(y)$
- 7 **return** α, ϕ^*

Probabilistic Inference for BNs: Variable Elimination

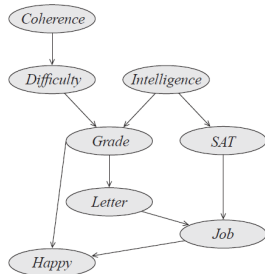


Figure 9.8 The Extended-Student Bayesian network

$$\begin{aligned}
 P(C, D, I, G, S, L, J, H) &= P(C)P(D | C)P(I)P(G | I, D)P(S | I) \\
 &\quad P(L | G)P(J | L, S)P(H | G, J) \\
 &= \phi_C(C)\phi_D(D, C)\phi_I(I)\phi_G(G, I, D)\phi_S(S, I) \\
 &\quad \phi_L(L, G)\phi_J(J, L, S)\phi_H(H, G, J).
 \end{aligned}$$

Step	Variable eliminated	Factors used	Variables involved	New factor
1'	C	$\phi_C(C), \phi_D(D, C)$	C, D	$\tau_1'(D)$
2'	D	$\phi_G[I = i^1](G, D), \phi_I[I = i^1](), \tau_1'(D)$	G, D	$\tau_2'(G)$
5'	G	$\tau_2'(G), \phi_L(L, G), \phi_H[H = h^0](G, J)$	G, L, J	$\tau_5'(L, J)$
6'	S	$\phi_S[I = i^1](S), \phi_J(J, L, S)$	J, L, S	$\tau_6'(J, L)$
7'	L	$\tau_6'(J, L), \tau_5'(J, L)$	J, L	$\tau_7'(J)$

Table 9.3 A run of sum-product variable elimination for $P(J, i^1, h^0)$

Beyond Variable Elimination

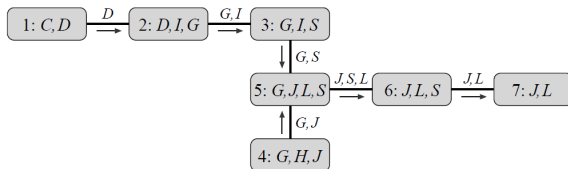


Figure 10.1 Cluster tree for the VE execution in table 9.1

- ▶ The execution of VE defines a cluster tree, a graphical flowchart of the factor-manipulation process:
 - ▶ Each node C_i in the tree is called a cluster and it contains the variables in $Scope(\psi_i)$.
 - ▶ The tree has an edge annotated with an arrow from C_i to C_j if the factor (a.k.a. message) τ_i is used in the computation of τ_j .
- ▶ Now, consider passing messages towards the cluster C_6 in order to compute $p(s)$. Notice the **reusing** of previously computed messages.
- ▶ Therefore, a clique tree is a data structure to perform **repeated** probabilistic inference efficiently.
- ▶ The process of building the cluster tree from a given DAG is known as compilation. You will learn more about cluster trees in the lab.

Contents

- ▶ Probabilistic Inference for BNs
 - ▶ Naive Solution
 - ▶ Variable Elimination
- ▶ Beyond Variable Elimination

Thank you